Worksheet 1.3 — Continuity & Limit Laws

Grade 12 Calculus and Vectors (MCV4U) · Unit 1: Rates of Change and Limits

The Big Ideas

Continuous at a: $f(a)$ defined, the limit exists, and they are equal. Discontinuities are removable, jump, or infinite.

- Three conditions: $f(a)$ defined, limit exists, $\lim = f(a)$.
- Removable (hole), jump (one-sided differ), infinite (asymptote).
- Limit laws split sums, products and quotients.

The Big Picture

Continuity

A function is **continuous** at a when $\lim_{x\to a} f(x) = f(a)$ no holes, jumps, or breaks in the graph there.

Limit laws

Limits distribute over sums, products, and quotients, so a complicated limit can be broken into simpler pieces.

Evaluating

For continuous functions a limit is found by direct substitution; only breaks require closer analysis.

Worked Examples

Example 1 — Verify continuity

Is $f(x) = x^2 + 1$ continuous at $x = 2$?

Solution. **Step 1 ($f(a)$ defined?):** $f(2) = 2^2 + 1 = 5$, which is defined.

Step 2 (limit exists?): f is a polynomial, so $\lim_{x\to 2}(x^2 + 1) = 2^2 + 1 = 5$ by direct substitution.

Step 3 (do they match?): $f(2) = 5$ and $\lim_{x\to 2} f(x) = 5$ — equal.

Conclusion: all three conditions hold, so f is continuous at $x = 2$.

Example 2 — Removable

Classify the discontinuity of $\dfrac{x^2 - 4}{x - 2}$ at $x = 2$.

Solution. **Step 1 ($f(a)$ defined?):** at $x = 2$ the denominator is 0, so $f(2)$ is undefined.

Step 2 (does the limit exist?): $\dfrac{x^2 - 4}{x - 2} = \dfrac{(x - 2)(x + 2)}{x - 2} = x + 2$ for $x \neq 2$, so $\lim_{x\to 2} f(x) = 4$, which exists.

Step 3 (compare): the limit exists (4) but $f(2)$ doesn't — a single missing point.

Conclusion: removable discontinuity (a hole at $(2, 4)$).

Example 3 — Infinite

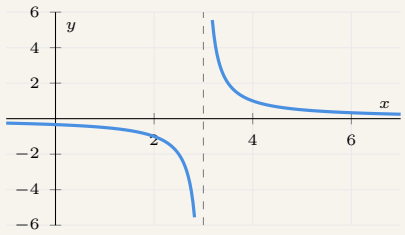
Classify the discontinuity of $\dfrac{1}{x-3}$ at $x=3$.

Solution. **Step 1 (right-hand limit):** as $x \to 3^+$, $x-3 \to 0^+$ (small positive), so $\dfrac{1}{x-3} \to +\infty$.

Step 2 (left-hand limit): as $x \to 3^-$, $x-3 \to 0^-$ (small negative), so $\dfrac{1}{x-3} \to -\infty$.

Step 3 (interpret): the function grows without bound on both sides — no finite limit, no hole to fill in.

Conclusion: infinite discontinuity (a vertical asymptote at $x=3$):



Example 4 — Jump

For $f(x) = \begin{cases} x, & x < 1 \\ x+2, & x \geq 1 \end{cases}$, what happens at $x=1$?

Solution. **Step 1 (left-hand limit):** for $x<1$, $f(x)=x$, so $\lim_{x\to 1^-} f(x) = 1$.

Step 2 (right-hand limit): for $x \geq 1$, $f(x)=x+2$, so $\lim_{x\to 1^+} f(x) = 1+2 = 3$.

Step 3 (compare): $1 \neq 3$ — the one-sided limits disagree, so the two-sided limit does not exist.

Conclusion: a jump discontinuity at $x=1$.

Example 5 — Make it continuous

Find k so $f(x) = \begin{cases} x^2, & x \leq 2 \\ kx, & x > 2 \end{cases}$ is continuous at 2.

Solution. **Step 1 (left piece at $x=2$):** $f(2) = 2^2 = 4$.

Step 2 (right piece's limit at $x=2$): $\lim_{x\to 2^+} kx = 2k$.

Step 3 (set them equal): for continuity the pieces must meet, so $2k = 4$.

Conclusion: $k=2$.

Example 6 — Where discontinuous

Where is $\dfrac{1}{x^2-9}$ discontinuous?

Solution. **Step 1 (find where the denominator is 0):** solve $x^2-9=0$.

Step 2 (factor): $(x-3)(x+3)=0$.

Step 3 (solve): $x=3$ or $x=-3$.

Conclusion: discontinuous (infinite, vertical asymptotes) at $x=3$ and $x=-3$.

Example 7 — Removable

Classify $\dfrac{x^2-1}{x-1}$ at $x=1$.

Solution. **Step 1 ($f(1)$ defined?):** the denominator is 0 at $x=1$, so $f(1)$ is undefined.

Step 2 (does the limit exist?): $\dfrac{x^2-1}{x-1} = \dfrac{(x-1)(x+1)}{x-1} = x+1$ for $x \neq 1$, so $\lim_{x\to 1} f(x) = 2$.

Conclusion: removable discontinuity (a hole at $(1,2)$).

Example 8 — Limit laws

$\lim\limits_{x\to 2}(3x^2 - x)$.

Solution. **Step 1 (split by the sum/difference law):** $\lim_{x\to 2} 3x^2 - \lim_{x\to 2} x$.

Step 2 (evaluate each piece by substitution, since both are continuous): $3(2)^2 - 2 = 3(4) - 2$.

Conclusion: $= 12 - 2 = 10$.

Example 9 — Domain & continuity

Where is $\sqrt{x}$ continuous?

Solution. **Step 1 (find the domain):** $\sqrt{x}$ is only defined for $x \geq 0$.

Step 2 (check continuity on that domain): for every $a \geq 0$, $\lim_{x\to a}\sqrt{x} = \sqrt{a} = f(a)$, so the three conditions hold throughout.

Conclusion: $\sqrt{x}$ is continuous on its whole domain, $x \geq 0$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Is $\dfrac{x+1}{x-2}$ continuous at $x = 2$?

Question 2

Classify the discontinuity of $\dfrac{x^2-1}{x-1}$ at $x = 1$.

Question 3

$\lim\limits_{x\to 3}(2x^2 - x)$?

Question 4

Find k: $\begin{cases} 3x, & x < 1 \\ x + k, & x \geq 1 \end{cases}$ continuous at 1.

Question 5

Where is $\dfrac{1}{x^2-4}$ discontinuous?

Question 6

Classify the discontinuity of $\dfrac{1}{x+5}$ at $x=-5$.

Question 7

Is $f(x)=x^3-2x$ continuous at $x=0$?

Question 8

$\lim\limits_{x\to 1}(4x^2+2x)$?

Question 9

Find k: $\begin{cases} x^2, & x\leq 3 \\ kx, & x>3 \end{cases}$ continuous at 3.

Question 10

Where is $\dfrac{x}{x^2 - x}$ discontinuous?

Question 11

Classify the discontinuity of $\dfrac{x^2 - 9}{x - 3}$ at $x = 3$.

Question 12

State the three conditions for continuity at a.

Question 13 — Challenge

For $\begin{cases} x^2 + 1, & x < 2 \\ ax + 1, & x \geq 2 \end{cases}$, find a making it continuous at 2.

Question Recap & Answer Key

Question Recap

1. Is $\dfrac{x+1}{x-2}$ continuous at $x=2$?
2. Classify the discontinuity of $\dfrac{x^2-1}{x-1}$ at $x=1$.
3. $\lim\limits_{x\to 3}(2x^2-x)$?
4. Find k: $\begin{cases} 3x, & x<1 \\ x+k, & x\geq 1 \end{cases}$ continuous at 1.
5. Where is $\dfrac{1}{x^2-4}$ discontinuous?
6. Classify the discontinuity of $\dfrac{1}{x+5}$ at $x=-5$.
7. Is $f(x)=x^3-2x$ continuous at $x=0$?
8. $\lim\limits_{x\to 1}(4x^2+2x)$?
9. Find k: $\begin{cases} x^2, & x\leq 3 \\ kx, & x>3 \end{cases}$ continuous at 3.
10. Where is $\dfrac{x}{x^2-x}$ discontinuous?
11. Classify the discontinuity of $\dfrac{x^2-9}{x-3}$ at $x=3$.
12. State the three conditions for continuity at a.
13. For $\begin{cases} x^2+1, & x<2 \\ ax+1, & x\geq 2 \end{cases}$, find a making it continuous at 2.

✓ Answer Key

1. no (infinite)
2. removable
3. 15
4. $k=2$
5. $x=\pm 2$
6. infinite
7. yes
8. 6
9. $k=3$
10. $x=0,1$
11. removable
12. $f(a)$ defined, limit exists, equal
13. $5=2a+1\Rightarrow a=2$